Here $A \times B$ is a subset of $\mathbb{R} \times \mathbb{R}$. The number of elements in $A \times B$ is the product of the number of elements in A and the number of elements in B, that is, $n(A \times B) = n(A)n(B)$, if A and B are finite. Further $n(A \times B \times C) = n(A)n(B)n(C)$, if A, B and C are finite.

It is easy to see that the following are the subsets of $\mathbb{R} \times \mathbb{R}$.

(i) $\{(x, 2x) : x \in \mathbb{R}\}$ (ii) $\{(x, x^2) : x \in \mathbb{R}\}$
(iii) $\{(x, \sqrt{x}) : x$ is a non-negative real number$\}$ (iv) $\{(x^2, x) : x \in \mathbb{R}\}$.
(v) $\{(x, -\sqrt{x}) : x$ is a non-negative real number$\}$

Example 1.1 Find the number of subsets of A if $A = \{x : x = 4n + 1,\ 2 \leq n \leq 5, n \in \mathbb{N}\}$.

Solution:
Clearly $A = \{x : x = 4n + 1,\ n = 2, 3, 4, 5\} = \{9, 13, 17, 21\}$.
Hence $n(A) = 4$. This implies that $n(\mathscr{P}(A)) = 2^4 = 16$.

Example 1.2 In a survey of 5000 persons in a town, it was found that 45% of the persons know Language $A, 25\%$ know Language $B, 10\%$ know Language $C, 5\%$ know Languages A and $B, 4\%$ know Languages B and C, and 4% know Languages A and C. If 3% of the persons know all the three Languages, find the number of persons who knows only Language A.

Solution:
This problem can be solved either by property of cardinality or by Venn diagram.

Cardinality: Given that $n(A) = 45\%$ of $5000 = 2250$
Similarly, $n(B) = 1250, n(C) = 500, n(A \cap B) = 250, n(B \cap C) = 200, n(C \cap A) = 200$ and $n(A \cap B \cap C) = 150$.
The number of persons who knows only Language A is

$$\begin{aligned} n(A \cap B' \cap C') &= n\{A \cap (B \cup C)'\} = n(A) - n\{A \cap (B \cup C)\}. \\ &= n(A) - n(A \cap B) - n(A \cap C) + n(A \cap B \cap C). \\ &= 2250 - 250 - 200 + 150 = 1950. \end{aligned}$$

Thus the required number of persons is 1950.

Venn diagram: We draw the Venn Diagram using percentage.

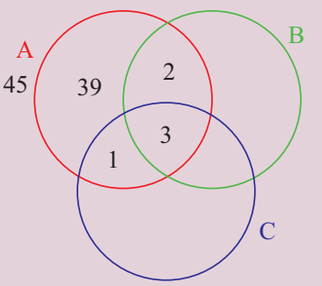


Figure 1.1

From Figure 1.1, the percentage of persons who knows only Language A is 39. Therefore, the required number of persons is $5000 \times \frac{39}{100} = 1950$.